Supplementary information

Vina docking-score matrices are strongly spectrally concentrated before, but not after, two-way centering

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Supplementary figures

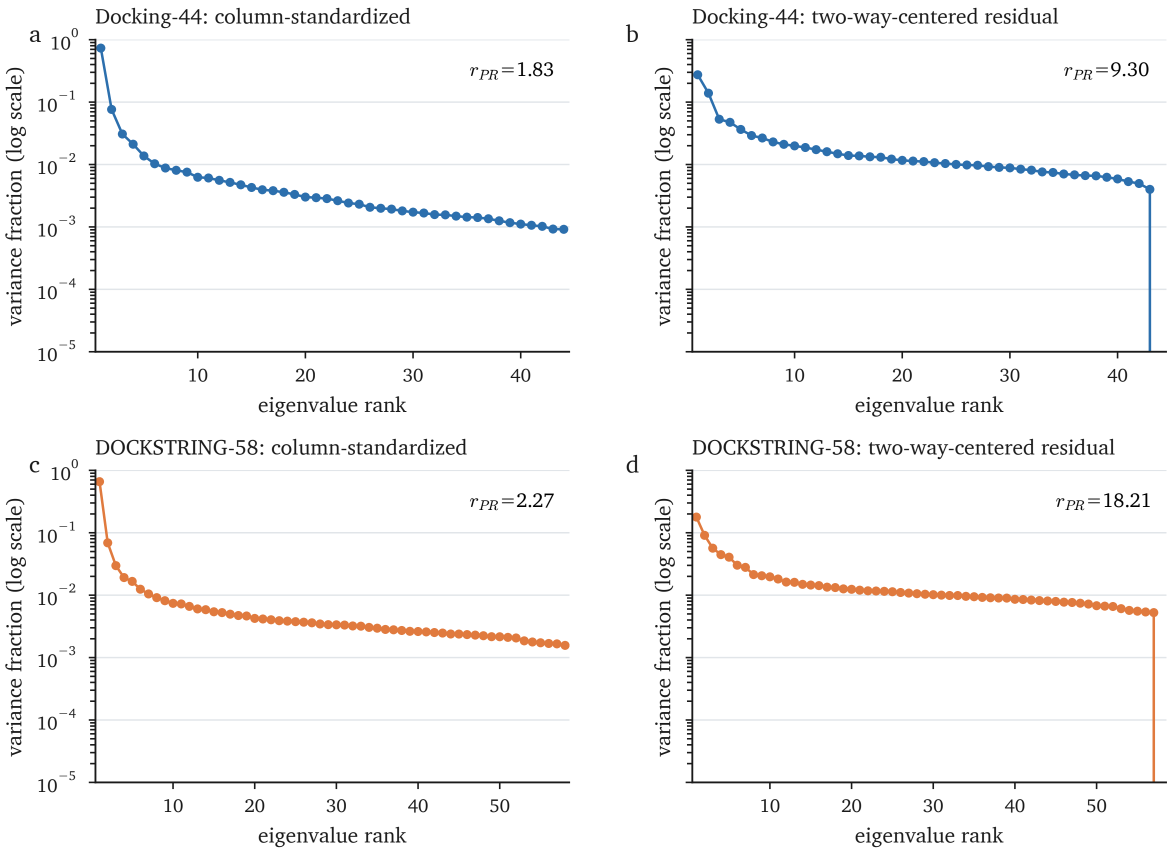

Figure S1: Full correlation-spectrum scree plots. a,b, Docking-44 column-standardized and two-way-centered residual spectra. **c,d**, DOCKSTRING-58 column-standardized and residual spectra. Eigenvalues are divided by the number of targets so that each spectrum sums to one. The ordinate is logarithmic. Double centering imposes one exact zero eigenvalue; values at zero fall below the plotting range.

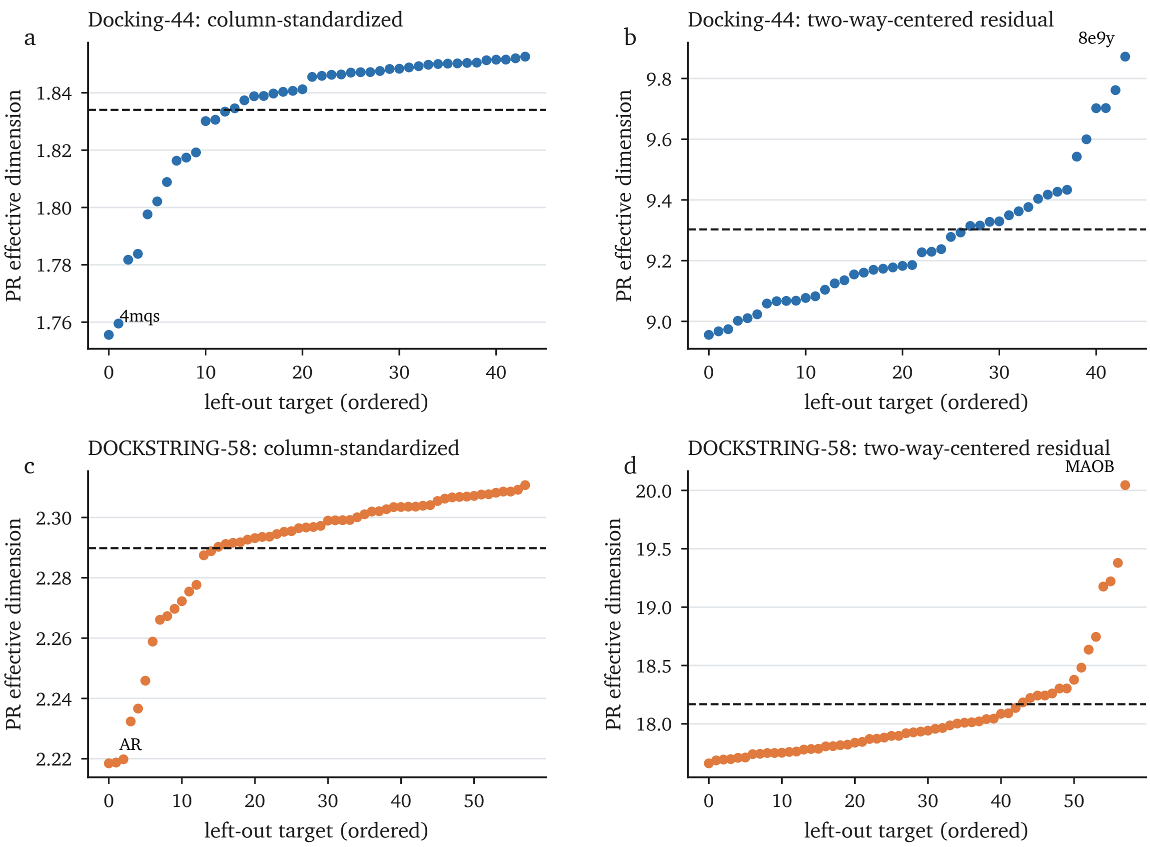

Figure S2: Leave-one-target-out composition sensitivity. Each point is the PR effective dimension after deleting one target; targets are ordered by the resulting value within each panel. Dashed lines mark the complete-panel estimate on the same ligand support. Labels identify the most influential deletion. These ranges describe the two observed target panels and are not confidence intervals for a target superpopulation.

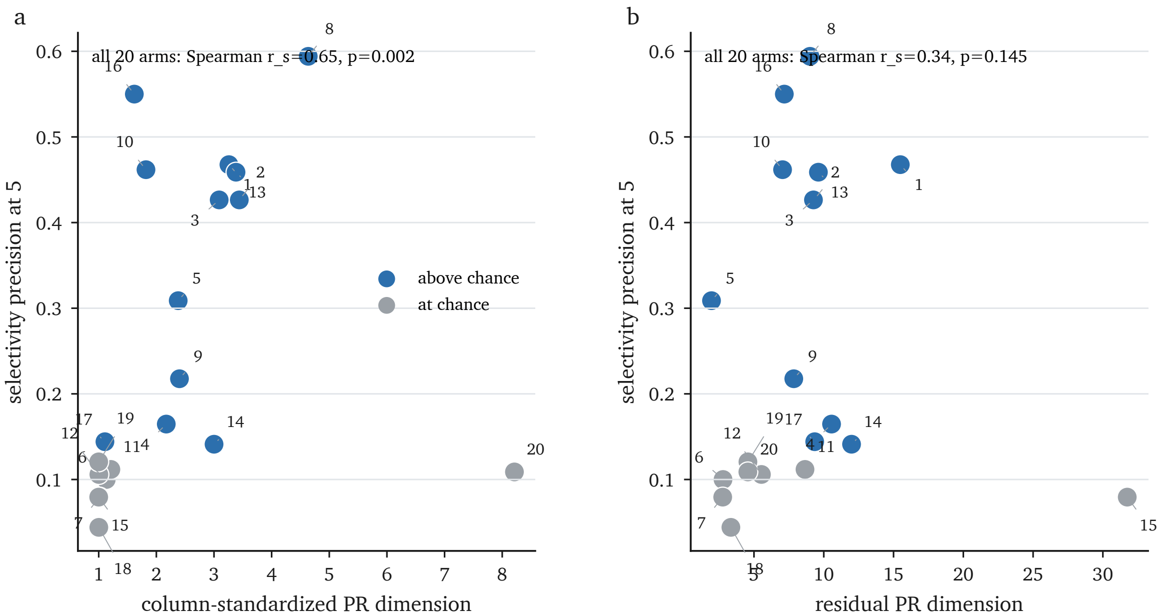

Figure S3: Exploratory DTI association on the complete 20-arm panel. Selectivity precision at five is plotted against the column-standardized PR effective dimension (**a**) and the two-way-centered residual PR effective dimension (**b**). Numbers identify arms in Table S5; blue denotes arms above the outcome-related chance criterion and grey denotes arms at chance. Correlations are descriptive because the arms share data, architectures and representations.

Table S1: Dataset and preprocessing summary. PR denotes participation-ratio effective dimension. Listed procedures quantify sensitivity within the observed chemical and target panels, not superpopulation uncertainty.

Matrix	scorer / source	N	P	missing	preprocessing	column-standardized PR	residual PR	entropy rank	uncertainty
Docking-44	Vina-GPU 2.0	12,651	44	1.466%	positive/source-censored scores to 0; target-mean imputation; column z-score	1.834	9.303	3.897	Butina-cluster bootstrap; target leave-one-out and subsampling
DOCKSTRING-58	AutoDock Vina	260,060	58	0.002%	positive scores to 0; complete rows; column z-score	2.266	18.206	6.206	Bemis-Murcko scaffold-cluster bootstrap on fixed chemical support; target jackknife
RF-Score/Vina-pose sensitivity	RF-Score v1	41	14	0.000%	complete common block; column z-score	1.416	6.681	2.158	ligand bootstrap; small fixed panel
RF-Score/Vinardo-pose sensitivity	RF-Score v1	35	20	0.000%	complete common block; column z-score	1.812	9.918	3.347	ligand bootstrap; small fixed panel
Matched docking	AutoDock Vina	74	6	0.000%	complete matched block; column z-score	1.827	3.646	2.605	same ligand-target support as exact-relation median ChEMBL block
Matched ChEMBL affinity	experimental pChEMBL	74	6	9.910%	exact relation; median over repeats; target-median imputation; column z-score	3.833	4.128	4.628	same observed support; assay and imputation sensitivities

Table S2: Docking-44 preprocessing and censoring sensitivity. Complete-case and target-exclusion results are not directly comparable because they change chemical or target support.

Surface	preprocessing	PR	note
Column standardized	target-mean imputation (primary)	1.834	same 12,651 ligands
Column standardized	target-median imputation	1.835	same support
Column standardized	complete cases	1.320	9,297 ligands; support changes
Column standardized	primary mean-imputed matrix restricted to complete rows	1.320	same 9,297-row support; equals complete case
Column standardized	pairwise-complete Pearson correlation	1.788	no matrix imputation
Column standardized	pairwise Spearman correlation	1.686	rank correlation
Column standardized	censored zeros treated as missing	1.829	111 cells; target-mean imputation
Column standardized	exclude ligands with censored zeros	1.814	12,544 ligands
Column standardized	exclude four most-censored targets	1.602	changes the target estimand
Residual	center then standardize (primary)	9.303	closed-form two-way centering
Residual	standardize then center	9.378	order sensitivity
Residual	robust scale then center	9.462	median/IQR scaling

Table S3: Experimental-assay, missing-data and repeatability controls. The noise scale was 0.493 pChEMBL, estimated from different-document repeats.

Analysis block	N	P	PR	note
ChEMBL all endpoints, frozen refetch A	212	22	6.846	already human-only; KNN; 52.9% missing
human binding Ki/Kd	175	19	5.248	assay type B; KNN
IC50/EC50	88	15	3.921	KNN
fully observed sub-block	31	6	3.608	no imputation
matched exact-relation median	74	6	3.833	same-support docking 1.83
matched human binding, all endpoints	47	6	3.510	same-support docking 1.87
matched human binding Ki/Kd	35	6	3.072	same-support docking 1.89
ChEMBL all endpoints, frozen refetch B, KNN5	212	22	5.721	different frozen extraction; 52.9% missing
212 x 22, MICE	212	22	8.592	mean across imputations
212 x 22, SoftImpute	212	22	2.644	low-rank matrix completion
212 x 22, PPCA-EM	212	22	7.283	probabilistic PCA
matched docking + calibrated noise	74	6	2.290	95% simulation interval 2.12–2.50
matched docking + 2.5x calibrated noise	74	6	3.877	approximately reaches experimental raw PR
matched docking, linear descriptor residual	74	6	2.806	seven RDKit descriptors; same support
matched docking, cross-fitted nonlinear residual	74	6	2.680	five-fold HistGBM; same support
matched docking residual, exact-relation support	74	6	3.646	two-way centered
matched experimental residual, exact relation median	74	6	4.128	two-way centered; primary curation
matched experimental residual, legacy most-potent	74	6	4.086	paired difference interval -0.24–0.93

Table S4: Target-panel and chemical-support sensitivities. Leave-one-target-out values describe composition sensitivity within each observed panel; they are not superpopulation confidence intervals.

Matrix	surface	full PR	LOTO min	LOTO median	LOTO max	target
Docking-44	column-standardized	1.834	1.756	1.846	1.853	4mqs
Docking-44	two-way-centered residual	9.303	8.956	9.206	9.872	8e9y
DOCKSTRING-58	column-standardized	2.266	2.218	2.297	2.311	AR
DOCKSTRING-58	two-way-centered residual	18.206	17.662	17.931	20.045	MAOB

Family-stratified target subsampling is shown in Fig. 1c,d. In five independent 100-permutation series, supra-null mode counts were 3 versus 7 in every series for Docking-44 and 4 versus 10 in every series for DOCKSTRING-58. On a seeded 15,000-molecule DOCKSTRING support, the molecule-bootstrap 95% interval was 2.230–2.296 and the Bemis–Murcko scaffold-cluster interval was 2.148–2.367.

Table S5: All 20 exploratory DTI scorer arms. Plot IDs identify points in Fig. S3. The above-chance flag is outcome-related and is not used to define this complete panel.

ID	arm	above chance	affinity r	column-standardized PR	residual PR	P@5	AUPRC
1	boltz_s3	yes	0.483	3.26	15.48	0.468	0.424
2	s1_readout	yes	0.553	3.44	9.30	0.426	0.423
3	chembert_davisoof	yes	0.609	3.09	9.26	0.426	0.379
4	chembert_kiba2davis	yes	0.154	2.17	10.56	0.165	0.208
5	conplex_kiba2davis	yes	0.378	2.38	1.96	0.309	0.311
6	deepdta_kiba2davis	no	0.072	1.13	2.78	0.100	0.161
7	dmpnn_kiba2davis	no	0.218	1.02	2.76	0.079	0.162
8	esm_target_davisoof	yes	0.744	4.63	9.01	0.594	0.545
9	esm_target_kiba2davis	yes	0.313	2.40	7.85	0.218	0.240
10	histgbm_davisoof	yes	0.617	1.82	7.05	0.462	0.412
11	histgbm_kiba2davis	no	0.196	1.21	8.65	0.112	0.178
12	kronrls_davisoof	no	0.279	1.00	5.53	0.106	0.143
13	mlp_davisoof	yes	0.618	3.38	9.62	0.459	0.393
14	mlp_kiba2davis	yes	0.152	3.00	11.99	0.141	0.203
15	morgan_cnn_kiba2davis	no	0.241	1.00	31.75	0.079	0.153
16	rf_davisoof	yes	0.638	1.62	7.17	0.550	0.460
17	rf_kiba2davis	yes	0.217	1.11	9.35	0.144	0.189
18	ridge_davisoof	no	0.304	1.00	3.34	0.044	0.116
19	ridge_kiba2davis	no	0.146	1.00	4.57	0.121	0.141
20	transformer_cnn_kiba2davis	no	0.083	8.21	4.56	0.109	0.151

Table S6: Provenance for the outcome-restricted 12-arm DTI sensitivity panel. DAVIS OOF denotes six-fold GroupKFold by target; KIBA transfer denotes zero-shot evaluation on DAVIS after KIBA training. Metrics are listed for all arms in Table S5.

ID	arm	representation / head	training and evaluation	caveat
1	Boltz-2 deployed affinity head	co-folded complex; deployed affinity score	foundation-model training; no DAVIS fitting in this analysis; external zero-shot DAVIS	foundation-model training corpus may contain related structures or affinities
2	Boltz-2 frozen representation + ridge	frozen pre-pairformer representation; RidgeCV	DAVIS training folds; six-fold unseen-target GroupKFold	shares Boltz-2 representation with boltz_s3
3	ChemBERTa + AAC, DAVIS OOF	ChemBERTa drug embedding + amino-acid composition; MLP	DAVIS training folds; six-fold unseen-target GroupKFold	paired with chembert_kiba2davis
4	ChemBERTa + AAC, KIBA to DAVIS	ChemBERTa drug embedding + amino-acid composition; MLP	KIBA; zero-shot transfer to DAVIS	paired with chembert_davisoof
5	ConPLex	protein language model + contrastive DTI score	pretrained on BindingDB; zero-shot DAVIS inference	possible training-data overlap with DAVIS proteins
8	Morgan + ESM-2, DAVIS OOF	Morgan drug features + ESM-2 target embedding; MLP	DAVIS training folds; six-fold unseen-target GroupKFold	paired with esm_target_kiba2davis
9	Morgan + ESM-2, KIBA to DAVIS	Morgan drug features + ESM-2 target embedding; MLP	KIBA; zero-shot transfer to DAVIS	paired with esm_target_davisoof
10	Morgan + AAC gradient boosting	Morgan + physicochemical drug features and amino-acid composition; HistGBM	DAVIS training folds; six-fold unseen-target GroupKFold	feature family overlaps several comparator arms
13	Morgan + AAC MLP, DAVIS OOF	Morgan + physicochemical drug features and amino-acid composition; MLP	DAVIS training folds; six-fold unseen-target GroupKFold	paired with mlp_kiba2davis
14	Morgan + AAC MLP, KIBA to DAVIS	Morgan + physicochemical drug features and amino-acid composition; MLP	KIBA; zero-shot transfer to DAVIS	paired with mlp_davisoof
16	Morgan + AAC random forest, DAVIS OOF	Morgan + physicochemical drug features and amino-acid composition; random forest	DAVIS training folds; six-fold unseen-target GroupKFold	paired with rf_kiba2davis
17	Morgan + AAC random forest, KIBA to DAVIS	Morgan + physicochemical drug features and amino-acid composition; random forest	KIBA; zero-shot transfer to DAVIS	paired with rf_davisoof

Table S7: Selection and estimator sensitivity of rank–selectivity associations. The 12-arm panel excludes eight chance-level arms on an outcome-related criterion. All coefficients are descriptive across dependent model arms.

Model panel	PR variable	endpoint	Pearson r	p	Spearman ρ	p	Kendall τ	p
12 above-chance	column-standardized PR	P@5	0.467	0.1261	0.347	0.2695	0.260	0.2426
12 above-chance	residual PR	P@5	-0.046	0.8877	-0.291	0.3593	-0.198	0.3716
12 above-chance	column-standardized PR	AUPRC	0.542	0.0685	0.462	0.1309	0.364	0.1160
12 above-chance	residual PR	AUPRC	-0.026	0.9354	-0.245	0.4433	-0.152	0.5452
all 20 scorers	column-standardized PR	P@5	0.309	0.1854	0.645	0.0021	0.483	0.0031
all 20 scorers	residual PR	P@5	0.039	0.8697	0.338	0.1452	0.243	0.1352
all 20 scorers	column-standardized PR	AUPRC	0.313	0.1791	0.660	0.0016	0.544	0.0008
all 20 scorers	residual PR	AUPRC	0.065	0.7866	0.373	0.1053	0.221	0.1859

For column-standardized PR versus P@5 in the 12-arm subset, the arm-resampling 95% interval was -0.39–0.89, leave-one-arm-out values ranged from 0.15 to 0.57, and the approximate two-sided 0.05 detectable absolute correlation was 0.58.

Additional methodological notes

Interpretation of PR effective dimension. For a P-column standardized matrix with target correlation matrix C,

$$r_{\mathrm{PR}} = \frac{P^2}{\mathrm{tr}(C^2)} = \frac{P}{1 + (P-1)\overline{r_{ij}^2}}.$$

The mean is over unique off-diagonal target correlations. The PR effective dimension therefore summarizes mean-squared inter-target correlation, while the plotted spectrum distinguishes a single dominant collective direction from several intermediate modes.

Limits of the experimental-noise calibration. The different-document repeatability scale includes technical variation, laboratory differences and assay-condition variation. It is deliberately conservative for a technical noise control, but it cannot represent target-specific systematic bias or all biological heterogeneity. The simulation therefore asks whether noise at the observed repeatability scale is sufficient to reproduce the experimental PR effective dimension; it is not a measurement-error correction of ChEMBL.

DTI model-panel interpretation. All 20 scorer arms are the primary exploratory panel in Table S5 and Fig. S3. The 12 arms in Table S6 share data, architectures and representations and were retained only as an explicitly outcome-restricted sensitivity. Table S7 compares both panels. The positive column-standardized-PR association in the all-20 panel and its absence in the restricted 12-arm panel show that the conclusion is selection- and estimator-dependent. In particular, `transformer_cnn_kiba2davis` had column-standardized PR dimension 8.21 at chance-level retrieval, while `morgan_cnn_kiba2davis` had residual PR dimension 31.75 at chance. High dimension is therefore not sufficient for useful selectivity, just as low dimension is not sufficient to diagnose failure. Reported correlations and nominal probabilities describe these panels only; they are not estimates of a population-level causal effect of dimensionality on selectivity.

For precision at five, we applied 20 iterations of Tukey-style median polish separately to predicted and experimental floor-excluded DAVIS matrices. Ligands with fewer than 12 co-observed targets were omitted. Experimental residuals at or above the within-ligand top decile, including ties, were labelled relevant; precision among the five largest predicted residuals was averaged over 68 eligible ligands. Eligible ligands had 13–402 measured targets (median 106), and 61 of 68 had an attainable precision ceiling of one.

Common-protocol parallel analysis. For each large docking matrix, the column-standardized and residual spectra were computed on the same 12,000-row subsample. Five hundred null spectra independently permuted each target column and then applied the corresponding transformation. The counts above the rank-wise 95th percentile null were 3 versus 7 for Docking-44 and 4 versus 10 for DOCKSTRING-58. Each result was unchanged in five independent 100-permutation series. Because centering changes the estimand and creates one exact linear constraint, each surface is compared with its own transformed null.